

Data Flow Diagram

Date	4 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Created a Data Flow Diagram (DFD) to show how data moves through system - between external entities, processes, and data stores.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A user outside the system who interacts with the application, such as a Farmer, Agricultural Researcher, or Policymaker , by providing soil and environmental data or receiving crop recommendations.
Rectangle with numbered header	Process	An operation that processes the input data, such as Receive User Input (1.0), Data Preprocessing (2.0), Crop Prediction using Machine Learning (3.0), or Display Crop Recommendation (4.0)
Rectangle (solid fill, no number)	Data Store	Stores the agricultural dataset and trained machine learning model used for crop prediction, such as the Crop Dataset and Trained ML Model
Labeled arrow	Data Flow	Represents the movement of data between entities, processes, and the data store, such as Soil Data, Environmental Parameters, Processed Data, and Crop Recommendation.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

